Prompt:

I want you to create LTL constraints from sentences. the set of LTL constraints that you are allowed to use are the following five:

- A -> F(B), which means that when A occurs, then B must eventually occur afterwards

- NOT(B) U A, which means that B must not occur until A has occured

- (A -> F(B)) AND (NOT(B) U A), which combines both of the above, so A must occur before B every time

- START -> A, which means that the first symbol must be A

- END -> A, which means that the last symbol must be A

you may switch A and B at will, where A and B are the symbols of the LTL constraint

There can be more than one constraint in a sentence

an example of the first constraint (response) would be "Once a claim has been approved, it should eventually be paid out." with "approve claim" as A and "pay out claim" as B, which translates to A -> F(B)

an example of the second constraint (precedence) would be: "a claim should be created before it can be approved" with "create claim" as A and "approve claim" as B, which translates to NOT(B) U A

an example of the third constraint(succession) would be "Once a claim has been rejected, it should not be paid out." with "reject claim" as A and "not pay out claim" as B, which translates to (A -> F(B)) AND (NOT(B) U A)

an example of the fourth constraint(initialize) would be “Every claim has to be created in the beginning.” with “create claim" as A, which translates to START -> A

an example of the fourth constraint(end) would be “After a claim has been paid, it is closed.” with “pay claim" as A, which translates to END -> A

Please wait for the sentences

additionally, please only use the following symbols to replace A and B in the sentences:

create claim

creation claim

approve claim

reject claim

payout

pay out claim

create case

arrange meeting

enter metadata

submit case

assign case id

agree date

examine patient

check x-ray risk

perform x-ray

perform surgery

apply cast

book ticket

payment ticket

check-in flight

receive invoice

decide invoice legit

receive all necessary approvals

confirm payment

receipt of an invoice

enter data

pay invoice

send request

give confirmation

register client data

provide room service

provide laundry service

leave

search report

bring in a defective computer

each of these activities

error is detected

register claim

Check recommendation

Mark as not ok

Mark as Ok

Receive request for new hardware

Analyze request

Deny request

Approve request

Receive offer

this is accepted

take care of invoice

receive customer request

need further review

submit office supply request

register requirement

reject request

ask to make change

select vendor

approval, rejection, change

submit vacation request

review order

inform customer

wait for x days

inform about delay

deliver package

Receive order

reserve part

ship bicycle to the customer

Receive amount request

Request approval

File expense

Receive expense report

check take place

supervisor approves

supervisor rejects

respond

approval claim

pay claim

case closed manually

manage case

hold meeting

inform

get local id

propose dates

perform x-ray risk

perform reposition

prescribe rehabilitation

remove cast

provide personal information

booking phase completion

money transfer

modify provided data

cancellation

enter all relevant data

close case

give approval

prepare proposal

proposal

bill

charge bill

put in a file together with the claim documentation

analyze the defect

test proper system functionality

execute repair activity

examine claim

send it back

Proceed with claim

inform employee via email

edit request

fill in form

decide if approved

receive hardware by IT department

Send back request to IT

receive request

return to petitioner/employee

go to purchase department

generate purchase order

send notification

return application

generate notification

order missing products

check again

process delivery

update system

assemble bicycle

Assess credit risk

Accept credit

Perform check

stop process

deposit of reimbursement

receives a notification

send email

please create LTL constraints for these 10 sentences:

A claim should be created, before it can be approved.

A claim must be created, before it is approved.

When a claim is created, it may be approved.

If a claim is created, it can be approved.

They have to create the claim, before they can approve it.

Only after a claim is created, it is possible to approve the claim.

If a claim is approved, then it must have been created first.

To approve a claim, it should be created first.

A claim can be approved, after it has been created.

Approving the claim can happen at anytime, unless it still needs to be created.

Response:

Here are the LTL constraints for the provided sentences:

NOT(approve claim) U create claim

NOT(approve claim) U create claim

create claim -> F(approve claim)

create claim -> F(approve claim)

NOT(approve claim) U create claim

NOT(approve claim) U create claim

NOT(approve claim) U create claim

NOT(approve claim) U create claim

create claim -> F(approve claim)

NOT(approve claim) U create claim

Prompt:

Now please these 10:

Creation of the claim is required, before it can be approved.

The creation of the claim is a condition for the approval of the claim.

Only after the creation of a claim, the approval of the claim is possible.

When approval of the claim occurs, creation of the claim must have preceded.

If approval of the claim occurs, then creation of the claim must have been carried out first.

The approval of the claim requires that the creation of the claim occurs before it.

The approval of the claim should be preceded by the creation of the claim.

A claim must be created, before it can be approved or rejected.

A claim should be approved before it can be paid out.

Once a claim has been approved, it should eventually be paid out.

Response:

Here are the LTL constraints for the provided sentences:

NOT(approve claim) U create claim

NOT(approve claim) U create claim

NOT(approve claim) U create claim

NOT(approve claim) U create claim

NOT(approve claim) U create claim

NOT(approve claim) U create claim

NOT(approve claim) U create claim

NOT(approve claim) U create claim, NOT(reject claim) U create claim

NOT(approve claim) U create claim

approve claim -> F(pay out claim)

Prompt:
now these 10:

Once a claim has been rejected, it should not be paid out.

Payout should end the process and therefore exclude everything.

A case can be closed manually at any time, unless we still need to pay out the claim.

The case can be managed at the union after it has been created.

After a case is created, management can and must arrange a meeting between the union case worker and the other case worker.

After a meeting is arranged it must be held.

The case worker should enter metadata on the case, inform about when he is available for participating in a meeting and then submit the case.

When a case is submitted it may get a local id at the union, but it should also subsequently be assigned a case id in LO.

When a case is submitted, LO should eventually propose dates.

Only after LO has assigned its case id it is possible to manage the case and for LO to propose dates.

Response:
Here are the LTL constraints for the provided sentences:

reject claim -> F(NOT(pay out claim)) AND (NOT(pay out claim) U reject claim)

pay out claim -> END

NOT(case closed manually) U (pay out claim)

create case -> F(manage case)

create case -> F(arrange meeting)

arrange meeting -> F(hold meeting)

enter metadata AND inform -> F(submit case)

submit case -> F(assign case id)

submit case -> F(propose dates)

NOT(manage case AND propose dates) U assign case id

Prompt:
now these 10:

Once a date has been agreed upon a meeting should eventually be held.

Every process instance starts by examining a patient.

If an x-ray is taken, then the x-ray risk must be checked before it

Performing a reposition, applying a cast and performing surgery require that x-rays are performed before.

If a surgery is performed, then prescription of rehabilitation occurs eventually after it.

After a cast is applied, eventually the cast is removed and vice versa, before every removal of a cast, a cast must be applied.

The process begins with the booking of the ticket

Personal information of passengers can be provided after the booking of the ticket

The payment of the ticket triggers the completion of the booking phase.

The payment of the ticket is eventually followed by the actual transfer of money

Response:
Here are the LTL constraints for the provided sentences:

agree date -> F(hold meeting)

START -> examine patient

NOT(perform x-ray) U check x-ray risk

NOT(perform reposition) U perform x-ray, NOT(apply cast) U perform x-ray, NOT(perform surgery) U perform x-ray

perform surgery -> F(prescribe rehabilitation)

apply cast -> F(remove cast) AND NOT(remove cast) U apply cast

START -> book ticket

book ticket -> F(provide personal information)

payment ticket -> F(booking phase completion)

payment ticket -> F(money transfer)

Prompt:
now these 10:

As long as the Check-in of the flight does not take place, the customers can still modify the provided data to change

After check-in of the flight, only the cancellation is admitted

The process starts when an invoice is received by the administration department.

If they decide that the invoice appears legit then they enter all relevant data into the system

When all necessary approvals have been received the invoice can be paid.

Once payment is confirmed, the invoice case should be closed.

The receipt of an invoice is required before the administration department can enter data or scan the invoice.

Enter data is required before any approval can be given.

Finally, we have to pay the invoice before we can confirm payment.

If a request is sent, then a proposal is expected to be prepared afterwards

Response:
Here are the LTL constraints for the provided sentences:

NOT(check-in flight) U modify provided data

check-in flight -> F(cancellation)

START -> receive invoice

decide invoice legit -> F(enter data)

receive all necessary approvals -> F(pay invoice)

confirm payment -> F(close case)

NOT(enter data) U receive invoice

NOT(give approval) U enter data

NOT(confirm payment) U pay invoice

send request -> F(prepare proposal)

Prompt:
now these 10:

A confirmation is supposed to be mandatorily given after the proposal, and vice-versa any proposal is expected to precede a confirmation

Every process instance has to start by registering client data

Every provided room service must be billed.

Every provided laundry service must be billed

When the client leaves, the bill must be charged

The Police Report related to the car accident is searched within the Police Report database and put in a file together with the Claim Documentation.

A customer brings in a defective computer and the manager analyzes the defect and gives back an invoice.

After each of these activities, the proper system functionality is tested.

If an error is detected, another arbitrary repair activity is executed, otherwise the repair is finished.

After a claim is registered, it is examined by a claims officer.

Response:
Here are the LTL constraints for the provided sentences:

prepare proposal -> F(give confirmation) AND NOT(give confirmation) U prepare proposal

START -> register client data

provide room service -> F(bill)

provide laundry service -> F(bill)

eave -> F(charge bill)

search report -> F(put in a file together with the claim documentation)

bring in a defective computer -> F(analyze the defect AND give back an invoice)

each of these activities -> F(test proper system functionality)

error is detected -> F(execute repair activity)

register claim -> F(examine claim)

Prompt:
now these 10:

This recommendation is then checked by a senior claims officer who may mark the claim as OK or Not OK.

If the claim is marked as Not OK, it is sent back to the claims officer and the recommendation is repeated.

If the claim is OK, the claim handling process proceeds.

The process of buying new hardware is started as soon as a request for new hardware is received by the IT department.

The next step is for the IT department to analyse the request and either approve or deny it.

If it is denied, the employee is informed via email about the denial and the process is ended.

However, if the request is approved it is edited and completed.

Once this offer is received, a form is filled in and sent to the management.

The management then analyses the request and decides if it is approved or not.

When this is accepted and the financial resources are found, the hardware is received by the IT department.

Response:
Here are the LTL constraints for the provided sentences:

Check recommendation -> F(Mark as Ok OR Mark as not Ok)

Mark as not Ok -> F(send it back AND Check recommendation)

Mark as Ok -> F(Proceed with claim)

START -> Receive request for new hardware

Receive request for new hardware -> F(Analyze request AND (Approve request OR Deny request))

Deny request -> F(inform employee via email)

Approve request -> F(edit request AND complete request)

Receive offer -> F(fill in form AND send to management)

send to management -> F(analyze request AND (decide if approved OR decide not approved))

this is accepted -> F(receive hardware by IT department)

Prompt:
now these 10:

After which the financial department will take care of the invoice and the processes is ended.

In the scenario where the management does not approve the request, it can send the request back to the IT department for rework, then the IT department has to go back to editing and completing a request.

If the management does not approve the request, it can also decide to deny the request after which the process is ended.

The loan approval process starts by receiving a customer request for a loan amount.

If the customer needs further review or the loan amount is for \$10,000 or more, the request is sent to the approver Web service.

The process of an Office Supply Request starts when any employee of the organization submits an office supply request.

Once the requirement is registered, the request is received by the immediate supervisor of the employee requesting the office supplies.

If the request is rejected, the process will end.

If the request is asked to make a change, then it is returned to the petitioner who can review the comments for the change request.

If the request is approved, it will go to the purchase department that will be in charge of making quotations (using a subprocess) and select a vendor.

Response:
Here are the LTL constraints for the provided sentences:

receive hardware by IT department -> F(take care of invoice)

decide not approved -> F(Send back request to IT AND edit request AND complete request)

decide not approved -> F(deny request)

START -> receive customer request

need further review -> F(send request to approver Web service)

START -> submit office supply request

register requirement -> F(receive request by supervisor)

reject request -> F(end process)

ask to make change -> F(return to petitioner/employee AND review comments)

approval -> F(go to purchase department AND select vendor)

Prompt:

After a vendor is selected and confirmed, the system will generate and send a purchase order and wait for the product to be delivered and the invoice received.

In any of the cases, approval, rejection or change required the system will send the user a notification.

The process starts when any employee of the organization submits a vacation request.

Once the requirement is registered, the request is received by the immediate supervisor of the employee requesting the vacation.

If the request is rejected, the application is returned to the applicant/employee who can review the rejection reasons.

If the request is approved, a notification is generated to the Human Resources Representative, who must complete the respective management procedures.

The process starts by reviewing the order.

If something is missing, the staff has to inform customer about a potential delay and order the missing products.

After waiting an average of x days for the delivery, the staff should check again if all products are on store.

The courier company informs the client about a potential delay until there are enough human resources to process the delivery.

Response:

Here are the LTL constraints for the provided sentences:

select vendor -> F(generate purchase order AND send purchase order AND wait for delivery AND receive invoice)

(approval OR rejection OR change) -> F(send notification)

START -> submit vacation request

register requirement -> F(receive request by supervisor)

reject request -> F(return application AND review rejection reasons)

approval -> F(generate notification AND complete management procedures)

START -> review order

something missing -> F(inform customer AND order missing products)

wait for x days -> F(check again)

not enough human resources -> F(inform about delay)

Prompt:
lastly, these 14:

Once the driver delivers the package, she should update the system so all stakeholders are notified.

Whenever the sales department receives an order, a new process instance is created.

If the storehouse has successfully reserved or back-ordered every item of the part list and the preparation activity has finished, the engineering department assembles the bicycle.

Afterwards, the sales department ships the bicycle to the customer and finishes the process instance.

If the requested amount is lower or equal to 1M$, the company assess the risk of the credit.

After the approval request, the credit could be accepted or rejected

If the requested amount is greater than 1M$, an approval must be requested.

If the requested amount is lower or equal to 1M$, the company assess the risk of the credit.

As soon as an employee files an expense report, the process is started.

Once the expense report is received, a check is performed to make sure the employee has an account.

If this check does not take place within 28 days, the process is stopped and the employee is sent a submit email.

First, the supervisor approves within 7 days, this will result in the deposit of the reimbursement directly to the employee's bank account.

Second, the supervisor rejects within 7 days, as a result the employee receives a notification through email that the request has been rejected.

Third, if it takes the supervisor more than 7 days to respond, an email is sent to the employee informing him/her that the approval is in progress and the part of the processes after a supervisor is asked for approval is repeated.

Response:

Here are the LTL constraints for the provided sentences:

deliver package -> F(update system AND notify stakeholders)

receive order -> F(create process instance)

(reserve items AND preparation activity finished) -> F(assemble bicycle)

assemble bicycle -> F(ship bicycle AND finish process instance)

amount <= 1M -> F(assess credit risk)

approval request -> F(accept credit OR reject credit)

amount > 1M -> F(request approval)

amount <= 1M -> F(assess credit risk)

file expense report -> F(start process)

receive expense report -> F(perform check)

¬(perform check within 28 days) -> F(stop process AND send submit email)

supervisor approves within 7 days -> F(deposit reimbursement)

supervisor rejects within 7 days -> F(send rejection notification)

supervisor response > 7 days -> F(send approval in progress email AND repeat approval process)